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Systemic Risk and Monetary Policy: The Haircut Gap Channel of the Lender of Last Resort

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1 Big picture

- **Question.** Does lender-of-last-resort (LOLR) policy affect systemic risk by changing banks' incentives to hold and pledge each other's bonds?
- **Core channel.** The paper studies the *haircut gap*: the difference between private-market repo haircuts and ECB collateral haircuts. A larger gap means that a bond is more attractive as central-bank collateral than as private-market collateral.
- **Main mechanism.** If a bank bond receives a low ECB haircut but a high private repo haircut, banks with ECB access can obtain more liquidity by pledging it to the ECB. This encourages banks to buy and pledge high-gap bank bonds.
- **Data.** The paper links security-level ECB collateral pledging, security holdings, ECB haircuts, private repo-market haircuts, bond characteristics, ratings, bank balance sheets, and market prices from January 2009 to March 2015.
- **Setting.** The euro area during and after the global financial crisis and sovereign debt crisis provides large variation in market stress, private haircuts, and the ECB's collateral policy.
- **Main pledging result.** A higher haircut gap increases bank pledging of eligible securities to the ECB. The effect is stronger for bank bonds than government bonds.
- **Bank-bond concentration result.** The effect is strongest for bank bonds issued by domestic banks, especially domestic periphery banks and systemically important banks.
- **Identification.** The paper uses both full-sample variation and institutional features of ECB haircuts: nonlinear rating notches and binding versus nonbinding downgrades around the A-/BBB+ cliff.
- **Holdings result.** High haircut gaps also raise banks' holdings of bank bonds. The effect is present for banks with LOLR access, not for nonbanks without LOLR access.
- **Issuance result.** High expected haircut gaps revive bank bond issuance: banks issue more high-gap bonds, at lower spreads and longer maturities.
- **Balance-sheet result.** Issuing banks expand liabilities and leverage; pledging banks increase ECB liquidity use and bank-bond holdings. The rise in bank-bond exposure is not offset by lower interbank lending.
- **Systemic-risk result.** Haircut-gap-induced interbank bond linkages increase comovement in bank bond prices, especially for domestic systemically important banks, in low-return periods, and during sovereign stress.
- **Economic takeaway.** LOLR policy can stabilize liquidity in a crisis but can also create incentives for banks to hold each other's risky debt, increasing interconnectedness and systemic risk.

2 Data and measurement

2.1 Haircut gaps

The central variable is the haircut gap:

$$\text{HaircutGap}_{s,t} = \text{PrivateHaircut}_{s,t} - \text{ECBHaircut}_{s,t}. \quad (1)$$

A positive value means that the private repo market demands a larger discount on security s than the ECB does.

Interpretation. A high haircut gap is an implicit collateral subsidy. It makes a security more useful for obtaining ECB funding than private repo funding.

2.2 Security pledging data

- The paper observes securities pledged by banks to the ECB in repo operations.
- Pledging is measured at the security–bank–month level.
- The main pledging outcome is

$$\frac{\text{ValuePledged}_{s,b,t}}{\text{ValueOutstanding}_s},$$

the value of security s pledged by bank b in month t , scaled by the total value outstanding of the security.

- The main sample contains millions of security–bank–month observations.

Interpretation. Scaling by security value outstanding makes the outcome comparable across bonds of very different sizes.

2.3 Security holdings data

- The holdings analysis uses security-level holdings across banking and nonbank sectors.
- The main holdings outcome is

$$\frac{\text{ValueHeld}_{s,c,t}}{\text{ValueOutstanding}_s},$$

where c denotes the country or sector of the holding institution.

- This allows the authors to compare banks with LOLR access to nonbanks without LOLR access.

Interpretation. The holdings evidence is important because pledging alone could reflect collateral sorting. Holdings show whether banks actually add high-gap bonds to their portfolios.

2.4 Bank bonds, government bonds, and domestic exposures

- The paper distinguishes bank-issued securities from government-issued securities.
- A bond is domestic if the pledging or holding bank and the issuing bank are located in the same country.
- The sample also separates core and periphery euro-area countries.
- The periphery group includes Italy, Spain, Portugal, Ireland, Cyprus, Malta, and Greece.

Interpretation. Domestic bank bonds are the key systemic-risk exposure. If banks hold and pledge each other's domestic debt, shocks can propagate through cross-holdings.

2.5 Systemic importance

- Systemic importance is measured with SRISK, following Brownlees and Engle.
- A bank is classified as systemically important if its SRISK ratio is in the top 10% of banks in its country.
- The paper studies whether high haircut gaps are especially associated with bonds issued by high-SRISK banks.

Interpretation. If the strongest effects involve systemically important issuing banks, the LOLR channel is more than a collateral-management issue; it directly affects systemic risk.

2.6 Issuance and balance-sheet data

- The issuance analysis studies newly issued bank bonds, their value, spread, and maturity.
- Because the haircut gap of a newly issued bond is not observed before issuance, the paper constructs an expected haircut gap using ECB rules and predicted private haircuts.
- Balance-sheet data track issuing banks and pledging banks, including bonds issued, ECB liquidity, bank-bond holdings, loans, deposits, liabilities, and leverage.

Interpretation. The issuance and balance-sheet tests ask whether haircut gaps merely reshuffle existing collateral or actually expand bank-bond finance and bank interconnectedness.

3 Models and empirical specifications

3.1 Baseline security-level specification

The generic specification is

$$y_{s,z,t} = \alpha_t + \alpha_s + \alpha_r + \alpha_z + \beta_1 \text{HaircutGap}_{s,t-1} + \beta_2 (\text{HaircutGap}_{s,t-1} \times H_{s,z}) + \gamma H_{s,z} + \delta X_{s,t-1} + \varepsilon_{s,z,t}. \quad (2)$$

Here z is either a pledging bank b or a holding country/sector c , α_t are time fixed effects, α_s are security fixed effects, α_r are rating fixed effects, and $X_{s,t-1}$ includes security-level controls.

Interpretation. The coefficient on the haircut gap compares the same security over time as its private and ECB haircuts diverge, while controlling for security, time, rating, and bank or country fixed effects.

3.2 Pledging of all bonds and bank bonds

The baseline pledging equation is

$$\frac{\text{ValuePledged}_{s,b,t}}{\text{ValueOutstanding}_s} = \alpha_t + \alpha_s + \alpha_r + \alpha_b + \beta_1 \text{HaircutGap}_{s,t-1} + \beta_2 (\text{HaircutGap}_{s,t-1} \times \text{BankBond}_s) + \delta X_{s,t-1} + \varepsilon_{s,b,t}. \quad (3)$$

Interpretation. β_1 captures the haircut-gap effect for government bonds; $\beta_1 + \beta_2$ captures the effect for bank bonds. A positive β_2 means the LOLR subsidy is more powerful for bank bonds.

3.3 Domestic, foreign, core, and periphery interactions

For bank bonds, the paper estimates

$$\frac{\text{ValuePledged}_{s,b,t}}{\text{ValueOutstanding}_s} = \alpha_t + \alpha_s + \alpha_r + \alpha_b + \beta_D (\text{HaircutGap}_{s,t-1} \times \text{Domestic}_{s,b}) + \beta_F (\text{HaircutGap}_{s,t-1} \times \text{Foreign}_{s,b}) + \delta X_{s,t-1} + \varepsilon_{s,b,t}. \quad (4)$$

It then expands the interactions by periphery and core pledging banks:

$$\frac{\text{ValuePledged}_{s,b,t}}{\text{ValueOutstanding}_s} = \alpha_t + \alpha_s + \alpha_r + \alpha_b + \beta_1 (\text{HaircutGap}_{s,t-1} \times \text{Periphery}_b \times \text{Domestic}_{s,b}) \quad (5)$$

$$+ \beta_2 (\text{HaircutGap}_{s,t-1} \times \text{Periphery}_b \times \text{Foreign}_{s,b}) \quad (6)$$

$$+ \beta_3 (\text{HaircutGap}_{s,t-1} \times \text{Core}_b \times \text{Domestic}_{s,b}) + \beta_4 (\text{HaircutGap}_{s,t-1} \times \text{Core}_b \times \text{Foreign}_{s,b}) + \delta X_{s,t-1} + \varepsilon_{s,b,t}. \quad (7)$$

Interpretation. This asks whether the haircut-gap channel concentrates risk inside domestic banking systems and whether the effect is stronger in stressed periphery countries.

3.4 Systemically important issuing banks

The paper adds interactions with SRISK:

$$\frac{\text{ValuePledged}_{s,b,t}}{\text{ValueOutstanding}_s} = \dots + \beta(\text{HaircutGap}_{s,t-1} \times \text{Domestic}_{s,b} \times \text{SRISK}_{s,t-1}) + \text{heta}(\text{HaircutGap}_{s,t-1} \times \text{SRISK}_{s,t-1}) + \varepsilon_{s,b,t}. \quad (8)$$

Interpretation. The key test is whether the haircut-gap incentive is strongest for bonds issued by banks whose distress would matter most for the financial system.

3.5 ECB institutional-rule identification

The institutional-rule tests exploit two features of ECB collateral policy.

- **Kinks and jumps:** private haircuts react to each rating notch, while ECB haircuts are step functions. This creates nonlinear changes in the haircut gap around rating notches.
- **Binding versus nonbinding downgrades:** around the A-/BBB+ cliff, an ECB haircut changes only when the first-best rating is downgraded. Downgrades that are nonbinding for ECB eligibility can still affect private market haircuts.

Interpretation. These tests reduce concern that unobserved security demand or broad risk trends drive the results. The identifying variation comes from institutional haircut rules rather than discretionary bank choice alone.

3.6 Holdings specification

For holdings, the paper estimates analogous specifications using

$$\frac{\text{ValueHeld}_{s,c,t}}{\text{ValueOutstanding}_s} \quad (9)$$

as the outcome. A simplified form is

$$\frac{\text{ValueHeld}_{s,c,t}}{\text{ValueOutstanding}_s} = \alpha_t + \alpha_s + \alpha_r + \alpha_c + \beta(\text{HaircutGap}_{s,t-1} \times \text{Domestic}_{s,c}) + \text{heta}X_{s,t-1} + \varepsilon_{s,c,t}. \quad (10)$$

Interpretation. This tests whether the collateral subsidy affects portfolio holdings, not only the choice of what collateral to pledge.

3.7 Banks versus nonbanks

To test whether LOLR access matters, the paper compares sectors:

$$\frac{\text{ValueHeld}_{s,i,t}}{\text{ValueOutstanding}_s} = \alpha_t + \alpha_s + \alpha_r + \alpha_i + \beta_1(\text{HaircutGap}_{s,t-1} \times \text{LOLRAccess}_i) + \beta_2(\text{HaircutGap}_{s,t-1} \times \text{NoLOLRAccess}_i) + \varepsilon_{s,i,t}. \quad (11)$$

Interpretation. If the channel is really LOLR-driven, the effect should appear for banks with ECB access but not for nonbank investors.

3.8 Bank-bond issuance

For newly issued bank bonds, the security-level issuance equation is

$$y_{s,t} = \alpha_t + \alpha_r + \alpha_c + \eta E_{t-1}(\text{HaircutGap}_{s,t}) + \eta X_{s,t} + \varepsilon_{s,t}, \quad (12)$$

where $y_{s,t}$ is log issue amount, bond spread, or log maturity.

At the bank level, the paper estimates

$$\log(\text{ValueOutstanding}_{b,t}) = \alpha_t + \alpha_b + \alpha_r + \beta \text{HaircutGap}_{b,t-1} + \varepsilon_{b,t}. \quad (13)$$

Interpretation. These equations test whether banks issue more bonds when those bonds are expected to be attractive ECB collateral.

3.9 Market-outcome correlation and systemic risk

The systemic-risk test uses correlations between pledging and issuing banks' market outcomes:

$$\text{Corr}(\text{BondPrices}_{i,b,t}) = \alpha_t + \alpha_{i,b} + \beta(\text{HaircutGap}_{i,t-1} \times \text{Domestic}_{i,b}) + \eta Z_{i,b,t-1} + \varepsilon_{i,b,t}. \quad (14)$$

The paper also interacts the haircut gap with SRISK, low market returns, and sovereign stress.

Interpretation. The question is whether LOLR-induced bank-bond linkages translate into more correlated market losses. This is the paper's key ex post systemic-risk evidence.

4 Identification and empirical design

4.1 What identifies the main effect

- The main variation is security-level, time-varying differences between private market haircuts and ECB haircuts.
- Security fixed effects absorb permanent differences across bonds.
- Time fixed effects absorb aggregate market stress and monetary policy conditions.
- Bank, country, sector, and rating fixed effects absorb stable cross-sectional differences.
- Lagged security controls address price, maturity, and rating-related confounds.

Interpretation. The baseline design asks whether the same security becomes more heavily pledged or held when its haircut gap rises.

4.2 Why ECB institutional rules matter

- ECB haircuts move in discrete rating buckets.
- Private repo haircuts respond more continuously to rating deterioration and market risk.
- This creates quasi-mechanical variation in the haircut gap.
- The binding/nonbinding downgrade design further isolates whether the ECB haircut actually changes after a downgrade.

Interpretation. These rule-based tests strengthen the interpretation that haircut gaps, not just endogenous risk demand, cause the pledging response.

4.3 Why the bank-versus-nonbank comparison matters

- Banks can use bonds as collateral at the ECB.
- Nonbanks cannot borrow from the LOLR.
- If high-gap bonds are attractive because of LOLR access, the response should be concentrated among banks.

Interpretation. This comparison is close to a placebo test. The same security should not become especially attractive to investors who cannot use the LOLR subsidy.

4.4 Why issuance and balance sheets matter

- Pledging and holding results show demand by banks.
- Issuance results show supply by bank bond issuers.
- Balance-sheet tests show whether bank-bond growth substitutes for other interbank linkages or expands the total network.

Interpretation. The paper’s strongest claim is systemic: the channel increases bank-to-bank linkages on net, not merely collateral substitution inside a fixed portfolio.

4.5 What the paper cannot fully prove

- Haircut gaps are not randomly assigned.
- ECB collateral rules provide powerful variation, but banks may still select securities based on unobserved characteristics.
- The paper focuses on the euro area and crisis-period institutional details.
- Market-outcome correlations indicate systemic-risk transmission but do not fully map into welfare or optimal LOLR design.

Interpretation. The evidence is strongest as a mechanism-based empirical case: many independent tests point to the same haircut-gap channel.

5 Tables and figures

The paper’s original figures and tables are directly cropped below. Adjacent items are grouped where possible to save space and make comparison faster.

5.1 Figures 1–3 and Table 1: where the collateral risk sits

Read: Figure 1:

- Banks are the largest holders of both government bonds and bank bonds.
- Bank bonds are especially concentrated in the banking sector.
- Domestic holdings dominate for bank-held bank bonds.

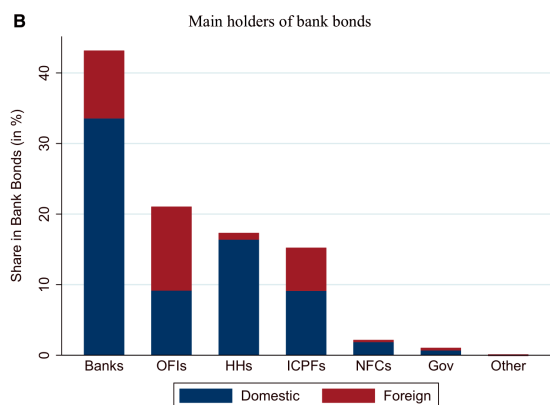
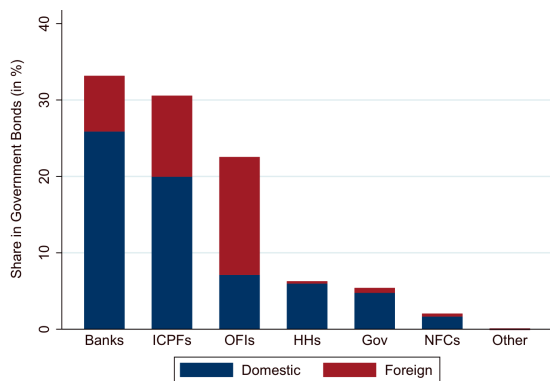
Read: Figure 2 and Figure 3:

- Bank-issued securities are a large part of ECB marketable collateral.
- Pledging of bank bonds changes strongly during the sovereign debt crisis.
- Newly issued bonds are pledged quickly: most pledging occurs within the first three months after issuance.

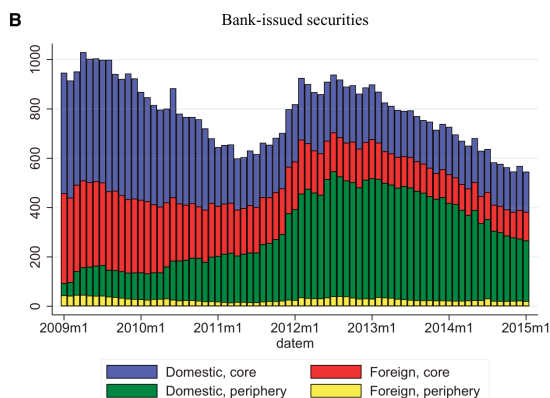
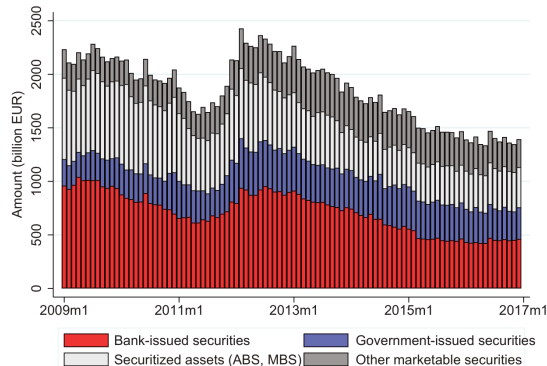
Read: Table 1:

- The main pledging sample has over 3.7 million observations.
- The average haircut gap is 6.06 percentage points.
- Private market haircuts average 11.98%, while ECB haircuts average 5.91%.
- The average value pledged is 5.16% of value outstanding.

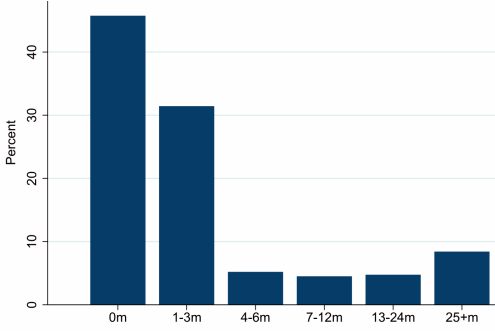
Economic message. The raw data show why the paper focuses on bank bonds: they are central to bank balance sheets and ECB collateral use, and haircut gaps create a direct incentive to pledge them.



(a) Figure 1: bond holdings by sector.



(b) Figure 2: ECB collateral pledging.



(a) Figure 3: timing of pledging.

		N	Mean	SD
<i>Main pledging sample: Security-bank-month level</i>				
Haircut gap	in percentage points	3,757,580	6.06	11.97
Private market haircut	in %	3,757,580	11.98	14.86
ECB haircut	in %	3,757,580	5.91	5.96
Value pledged	in % of value outstanding	3,757,580	5.16	18.50
Domestic	1=yes, 0=no	3,757,580	0.60	0.40
Periphery	1=yes, 0=no	3,757,580	0.23	0.42
SRISK	1=yes, 0=no	2,586,886	0.27	0.44
<i>Additional security characteristics: Security-month level</i>				
Haircut gap	in percentage points	477,104	5.19	10.00
Private market haircut	in %	477,104	11.79	12.53
ECB haircut	in %	477,104	6.60	5.27
Value pledged	in % of value outstanding	477,104	41.40	39.40
Residual maturity	in years	477,104	3.56	4.11
Bond price	log	477,104	-0.008	0.112
Rating	numerical scale	477,104	4.1	2.7
Issuance	in mil EU	8,245	445.09	1,273.63
<i>Holding sample: Security-country-quarter level</i>				
Haircut gap	in percentage points	184,565	4.52	7.6
Value held	in % of value outstanding	184,565	23.38	32.6
<i>Collapsed pledging sample: Security-country-quarter level</i>				
Haircut gap	in percentage points	163,993	6.09	12.0
Value pledged	in % of value outstanding	163,993	22.33	35.0

This table show the summary statistics of key variables for the sample period January 2009 to March 2015

(b) Table 1: summary statistics.

5.2 Figures 4–5: haircut gaps over time and by rating

Read: Figure 4:

- Before the sovereign debt crisis, core and periphery haircut gaps are relatively close.
- After 2010, the periphery haircut gap jumps sharply and remains high.
- Core haircut gaps stay low and stable.

Read: Figure 5:

- Private market haircuts rise gradually with rating deterioration.
- ECB haircuts are stepwise and jump at specific rating buckets.
- The haircut gap becomes much larger for lower-rated bonds.

Economic message. The figures motivate the identification strategy: the haircut gap is driven by crisis-time divergence between market haircuts and ECB collateral rules, especially in stressed periphery assets and lower rating notches.

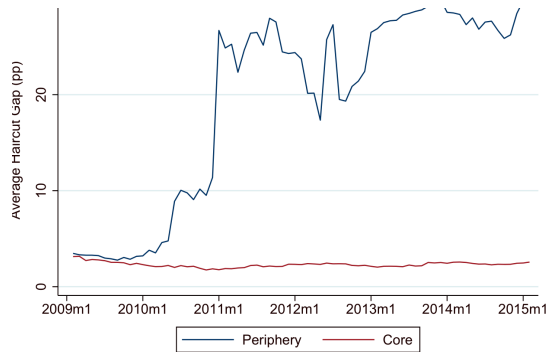
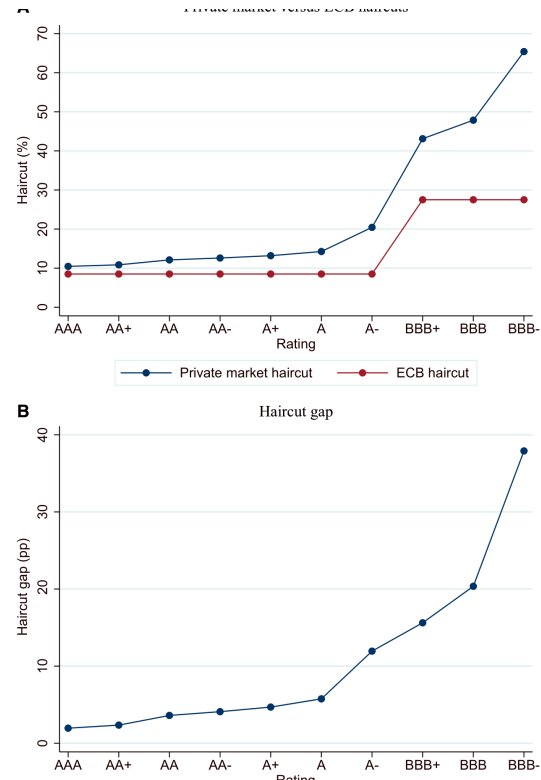


Figure 4: Average haircut gap for securities issued in core and periphery

(a) Figure 4: core versus periphery haircut gaps.



(b) Figure 5: haircuts and haircut gap by rating.

5.3 Tables 2–3: pledging behavior and ECB institutional rules

Read: Table 2:

- Haircut gaps significantly increase pledging.
- The bank-bond interaction is positive: the haircut-gap effect is stronger for bank bonds than for government bonds.
- Domestic bank bonds receive much stronger responses than foreign bank bonds.
- The strongest estimates come from domestic bonds pledged by periphery banks and from bonds of systemically important issuing banks.
- A one-standard-deviation increase in haircut gap is associated with about a 1.9% increase in bank-bond pledging relative to the mean.

Read: Table 3:

- The ECB rule-based tests confirm the full-sample findings.
- Kinks and jumps in the rating schedule generate positive effects for domestic bank bonds.
- Binding-downgrade tests also show stronger effects for domestic periphery bonds and SRISK banks.

Economic message. The baseline and institutional-rule evidence point to the same mechanism: favorable LOLR collateral treatment induces banks to pledge more risky, domestically connected bank debt.

Table 2
LOLR and banks' pledging behavior

	Value pledged _{b,t} Value outstanding _t				
	All bonds		Bank bonds		
	(1)	(2)	(3)	(4)	(5)
HaircutGap _{t-1}	0.00369*** (0.000308)	0.00295*** (0.000347)	0.00229*** (0.000594)	0.00114 (0.00167)	
HaircutGap _{t-1} × Bank Bond _{it}	0.00434*** (0.000780)	0.00506*** (0.000788)	0.00207*** (0.000848)		
HaircutGap _{t-1} × Domestic _{it}			0.0131*** (0.00147)	0.00899*** (0.00344)	
HaircutGap _{t-1} × Foreign _{it}			-0.0004 (0.00135)		
HaircutGap _{t-1} × Periphery _{it} × Domestic _{it}			0.0168*** (0.00032)		
HaircutGap _{t-1} × Periphery _{it} × Foreign _{it}			0.00175 (0.00038)		
HaircutGap _{t-1} × Corp _{it} × Domestic _{it}			0.000382 (0.00043)		
HaircutGap _{t-1} × Corp _{it} × Foreign _{it}			-0.00008 (0.00146)		
HaircutGap _{t-1} × Domestic _{it} × SRISK _{it-1}			0.0224*** (0.00085)		
HaircutGap _{t-1} × SRISK _{it-1}			-0.0109*** (0.00222)		
Controls	No	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Pledging bank FE	Yes	Yes	Yes	Yes	Yes
RSIN FE	Yes	Yes	Yes	Yes	Yes
Rating group FE	No	No	Yes	Yes	Yes
N	3,757,583	3,757,580	3,757,580	2,662,362	2,586,886
R ²	.867	.867	.867	.869	.872

This table presents the analysis of bond pledging. Columns 1-3 report coefficients from the regressions focused on pledging of government versus bank bonds, as described in Equation (1). The categorical variable Bank Bond takes the value of one if the bond is issued by a bank, and zero if it is issued by the government. Columns 4-5 present coefficients from the analysis of bank bond pledging, as described in Equations (4), (5), and (6). The table reports the main coefficients of interest, but the regressions estimate all coefficients as specified in these equations. A security is considered to be Domestic if the pledging and issuing banks have the same country of location, and as Foreign otherwise. SRISK takes the value of one if the issuing bank's SRISK ratio is in the top 10% of the SRISK ratios of all banks in the respective country, and zero otherwise. Controls include log values of security residual maturity and price. Standard errors are clustered at the security and time levels in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 3
LOLR and banks' pledging behavior: Analysis using ECB institutional rules

	Value pledged _{b,t} Value outstanding _t				
	Strategy 1: Kinks and jumps		Strategy 2: Binding downgrades		
	(1)	(2)	(3)	(4)	(5)
HaircutGap _{t-1} × Domestic _{it}	0.016*** (0.0057)		0.002 (0.0112)	0.017*** (0.0063)	-0.011 (0.0242)
HaircutGap _{t-1} × Foreign _{it}	-0.014*** (0.0051)			-0.006 (0.0057)	
HaircutGap _{t-1} × Periphery _{it} × Domestic _{it}		0.016*** (0.006)		0.028*** (0.0071)	
HaircutGap _{t-1} × Periphery _{it} × Foreign _{it}		-0.009 (0.0067)		-0.018 (0.0113)	
HaircutGap _{t-1} × Corp _{it} × Domestic _{it}		0.006 (0.0138)		0.004 (0.0061)	
HaircutGap _{t-1} × Corp _{it} × Foreign _{it}		-0.013** (0.0055)		-0.088*** (0.0190)	
HaircutGap _{t-1} × Domestic _{it} × SRISK _{it-1}			0.010*** (0.0171)	0.045* (0.0255)	
HaircutGap _{t-1} × SRISK _{it-1}			-0.023** (0.0096)	0.009 (0.0156)	
HaircutGap _{t-1}			0.0001 (0.0053)	-0.015 (0.0153)	
Controls	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Pledging bank FE	Yes	Yes	Yes	Yes	Yes
RSIN FE	Yes	Yes	Yes	Yes	Yes
Rating group FE	Yes	Yes	Yes	Yes	Yes
N	137,587	137,587	129,911	42,130	33,242
R ²	.891	.891	.898	.925	.951

This table presents the coefficients from the analysis of bank bonds based on identification strategies using the ECB institutional rules, as described in Equations (1), (5), and (6). Strategy 1: Kinks and jumps exploits nonlinearities in the haircut gap with respect to the rating. Strategy 2: Binding Downgrades uses the fact that, contrary to the private markets, the ECB considers only the best rating across the four rating agencies. The table reports the main coefficients of interest, but the regressions estimate all coefficients as specified in these equations. A security is deemed to be Domestic if the pledging and issuing banks have the same country of location, and as Foreign otherwise. SRISK takes the value of one if the issuing bank's SRISK ratio is in the top 10% of the SRISK ratios of all banks in the respective country, and zero otherwise. Controls include log values of security residual maturity and price. The dash refers to the fact that the fixed effects are not applicable given the scope of the analysis. Standard errors are clustered at the security and time levels in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

5.4 Tables 4–5: motives and robustness

Read: Table 4:

- The effect of high haircut gaps is stronger for banks with lower liquidity ratios.
- The effect is also stronger for lower-rated banks.
- This supports both liquidity-constraint and expected-subsidy-maximization motives.

Read: Table 5:

- The domestic versus foreign result remains stable under many fixed-effect structures.
- The SRISK result also remains robust.
- The table rules out several alternative explanations based on stable bank–issuer relationships or common country risk.

Economic message. Banks most likely to value cheap central-bank liquidity respond most strongly, and the core pledging results survive demanding robustness checks.

5.5 Tables 6–7: holdings and the role of LOLR access

Read: Table 6:

- Haircut gaps increase bank holdings of bank bonds.
- The holdings increase is concentrated in domestic bank bonds.
- The effect is stronger for periphery domestic bonds and for bonds issued by systemically important banks.
- The pledging and holdings margins are similar in economic magnitude.

Read: Table 7:

- Banks with LOLR access increase holdings of high-gap bank bonds.
- Nonbanks without LOLR access do not show the same response.
- For government bonds, the effect is smaller.

Table 4
Motives: Liquidity constraints versus expected subsidy maximization

	Value pledged _{s,t} Value outstanding _s			Value pledged _{s,c,t} Value outstanding _s		
	(1)	(2)	(3)	(4)	(5)	(6)
HaircutGap _{s,t-1}	0.043*** (0.0058)	-0.000122 (0.00291)	0.00333 (0.00201)			
HaircutGap _{s,t-1} × Liquidity Ratio _{s,t-1}	-0.312*** (0.0301)		-0.489*** (0.0429)	-0.361*** (0.0314)	-0.561*** (0.0457)	
HaircutGap _{s,t-1} × Rating _{s,t-1}		0.00237*** (0.000705)	0.00982*** (0.00100)		0.00401*** (0.000824)	0.0132*** (0.00122)
HighHaircutGap _{s,t-1} × value held _{s,c,t} value outstanding _s						0.901*** (0.0428)
LowHaircutGap _{s,t-1} × value held _{s,c,t} value outstanding _s						0.493*** (0.0033)
Controls	Yes	Yes	Yes	–	–	–
Time FE	Yes	Yes	Yes	–	–	–
Pledging bank FE	Yes	Yes	Yes	–	–	–
ISIN FE	Yes	Yes	Yes	–	–	–
Rating group FE	Yes	Yes	Yes	–	–	–
Pledging bank-Time FE	No	No	No	Yes	Yes	Yes
ISIN-Time FE	No	No	No	Yes	Yes	Yes
N	449,682	386,609	388,609	389,505	330,760	330,760
R ²	.855	.848	.848	.459	.453	.454

This table presents the coefficients from the regressions related to the pledging of bank bonds, as described in Equation (2) in columns 1-6, and as described in Equation (7) in column 7. Liquidity ratio is computed as a share of pledging bank's liquid assets to total assets. Rating represents a numerical bank rating where higher values denote worse rating quality. Liquidity ratio and rating are de-meaned. HighHaircutGap_{s,t-1} (LowHaircutGap_{s,t-1}) takes value of one when haircut gap is in the upper (bottom) quintile of haircut gap realizations, and zero otherwise. Controls include log values of security residual maturity and price. The dash refers to the fact that the fixed effects are not applicable as they are nested in different (higher-order) fixed effects. Standard errors are clustered at the security and time level in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

A. Domestic versus foreign

	Value pledged _{s,b,t} Value outstanding _s						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
HaircutGap _{s,t-1}	0.015*** (0.0024)	0.010*** (0.0028)	0.018*** (0.0030)	0.016*** (0.0024)	0.018*** (0.0027)	0.014*** (0.0030)	0.012*** (0.0029)
× Domestic _{s,b}	-0.0903 (0.0014)	0.0026* (0.0015)	-0.0015 (0.0016)	-0.0005 (0.0015)	0.0026 (0.0018)		
N	2,662,362	2,621,635	2,661,270	2,662,332	2,617,669	2,483,341	2,468,470
R ²	.869	.902	.936	.871	.873	.900	.941

B. SRISK

	Value pledged _{s,b,t} Value outstanding _s						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
HaircutGap _{s,t-1}	0.023*** (0.0039)	0.022*** (0.0043)	0.021*** (0.0048)	0.022*** (0.0039)	0.013*** (0.0045)	0.022*** (0.0055)	0.012*** (0.0053)
× Dom _{s,b} × SRISK _{s,t-1}	0.009*** (0.0034)	0.006 (0.0041)	0.017*** (0.0042)	0.010*** (0.0035)	0.014*** (0.0039)	0.011*** (0.0045)	0.012*** (0.0044)
× Domestic _{s,b}	-0.010*** (0.0022)	-0.010*** (0.0024)	-0.010*** (0.0027)	-0.012*** (0.0024)	-0.009*** (0.0032)		
HaircutGap _{s,t-1} × SRISK _{s,t-1}	0.001 (0.0017)	0.002 (0.0019)	-0.002 (0.0019)	0.001 (0.0018)	0.002 (0.0024)		
N	2,586,886	2,586,709	2,585,871	2,586,856	2,582,796	2,412,072	2,397,216
R ²	.872	.904	.938	.874	.876	.901	.942

Controls and fixed effects used in both panels

Controls	Yes ^a	Yes	Yes	Yes	Yes	–	–
Rating group FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time	Yes	Yes	Yes	–	–	–	–
ISIN FE	Yes	–	–	Yes	–	–	–
Pledging bank FE	Yes	–	–	Yes	Yes	Yes	–
Pled. bank-iss. bank FE	No	Yes	–	No	No	No	No
Pled. bank-ISIN FE	No	No	Yes	No	No	No	No
Iss. country-Time FE	No	No	No	Yes	–	–	–
Iss. bank-Time FE	No	No	No	No	Yes	–	–
ISIN-Time FE	No	No	No	No	No	Yes	Yes
Pled. bank-Time FE	No	No	No	No	No	No	Yes

This table presents the robustness to the coefficients from the regressions related to the pledging of bank bonds, as described in Equations (4) and (6). A security is considered to be Domestic if the pledging and issuing banks have the same country of location, and as Foreign otherwise. SRISK takes the value of one if the issuing bank's SRISK ratio is in the top 10% of the SRISK ratios of all banks in the respective country, and zero otherwise. Baseline controls include log values of security residual maturity and price. Superscript *a* includes further controls for the share of pledging of sovereign bonds. The dash refers to the fact that the fixed effects are not applicable as they are nested in different (higher-order) fixed effects. Standard errors are clustered at the security and time levels in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Economic message. The holdings evidence rules out a pure collateral-sorting story. LOLR access changes banks' portfolios, not just the set of assets they pledge.

	Value held _{s,c,t} Value outstanding _s			Value pledged _{s,c,t} Value outstanding _s	
	All bonds	Bank bonds		Bank bonds	
	(1)	(2)	(3)	(4)	(5)
HaircutGap _{s,t-1}	0.012*** (0.003)			-0.028*** (0.007)	
HaircutGap _{s,t-1} × Bank Bonds _s	0.016*** (0.006)				
HaircutGap _{s,t-1} × Domestic _{s,c}		0.074*** (0.014)		0.121*** (0.020)	0.043*** (0.014)
HaircutGap _{s,t-1} × Foreign _{s,c}		-0.006 (0.007)			0.006 (0.007)
HaircutGap _{s,t-1} × Periphery _c × Domestic _{s,c}			0.099*** (0.025)		
HaircutGap _{s,t-1} × Periphery _c × Foreign _{s,c}			-0.034*** (0.007)		
HaircutGap _{s,t-1} × Core _c × Domestic _{s,c}			0.056 (0.106)		
HaircutGap _{s,t-1} × Core _c × Foreign _{s,c}			-0.0351 (0.025)		
HaircutGap _{s,t-1} × Domestic _{s,c} × SRISK _{s,t-1}				0.155*** (0.026)	
HaircutGap _{s,t-1} × SRISK _{b,t-1}				-0.072*** (0.009)	
Controls	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Holding bank country FE	Yes	Yes	Yes	Yes	Yes
ISIN FE	Yes	Yes	Yes	Yes	Yes
Rating group FE	Yes	Yes	Yes	Yes	Yes
N	255,450	184,565	184,565	106,925	163,993
R ²	.875	.905	.974	.888	.883

This table presents the coefficients from the regressions related to bond holding, as described in Equation (2). Column 1 presents the results for both bank and government bonds, and columns 2-5 use only bank bonds. Bank Bond takes the value of one if the bond is issued by a bank, and zero if it is issued by the government. A security is considered to be Domestic if the pledging and issuing banks have the same country of location, and as Foreign otherwise. SRISK takes the value of one if the issuing bank's SRISK ratio is in the top 10% of the SRISK ratios of all banks in the respective country, and zero otherwise. Controls include log values of security residual maturity and price. Column 4 reports the main coefficients of interest but the regressions estimate all coefficients, including remaining interaction terms and controls. Column 5 presents the result of the pledging analysis at the identical source of data granularity as the holding estimates reported in column 2. We find that the two margins are aligned also in terms of magnitudes. A one-standard-deviation increase in the haircut gap is associated with a 2.4% increase in holding and 2.3% increase in pledging of domestic bank bonds compared to the mean value held/pledged as a share of the value outstanding at the country level. Standard errors are clustered at the security and time levels in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

(a) Table 6: haircut gap and security holdings.

	Value held _{s,l,c,t} Value outstanding _s			
	Bank bonds			Gov. bonds
	(1)	(2)	(3)	(4)
HaircutGap _{s,t-1}	0.0480*** (0.0062)	0.046*** (0.0062)	0.0435*** (0.0066)	0.0128*** (0.0020)
× LOLR access _s	-0.0105*** (0.0027)	-0.0125*** (0.0028)	-0.0086*** (0.0032)	0.00196*** (0.0008)
× No LOLR access _s				
Controls	No	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
ISIN FE	Yes	Yes	Yes	Yes
Sector-Country FE	Yes	Yes	Yes	Yes
Rating group FE	No	No	Yes	Yes
N	861,644	861,644	849,884	461,051
R ²	0.67	0.67	0.67	0.29

This table presents the coefficients from the regressions related to the holdings of bonds across sectors, as described in Equation (8). LOLR access denotes the banking sector. No LOLR access denotes other sectors (i.e., pension funds, mutual funds, insurance companies) that cannot borrow from the LOLR facilities. Controls include log values of security residual maturity and price. Standard errors are clustered at the security and time level in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

(b) Table 7: bank versus nonbank holdings.

5.6 Table 8 and Figure 6: bank-bond issuance

Read: Table 8:

- A higher expected haircut gap increases the value of newly issued bank bonds.
- It lowers issuance spreads and increases maturity.
- A one-standard-deviation increase in the haircut gap is associated with a 15.9% increase in value issued, a 50 bp lower spread, and a 43% longer maturity.
- At the bank level, a higher haircut gap raises total bank-bond debt outstanding by about 7.5% to 10.7%.

Read: Figure 6:

- Bank-bond issuance rises strongly in the periphery when haircut gaps are high.
- The timing visually matches the paper’s issuance regressions.

Economic message. The haircut-gap channel affects the supply side. Banks issue securities that other banks can use as attractive ECB collateral.

A. Security-level analysis

	log(issued _{s,t})		Spread _{s,t}		log(maturity _{s,t})	
	(1)	(2)	(3)	(4)	(5)	(6)
$E_{t-1}(\text{HaircutGap}_{s,t})$	1.85*** (0.603)	1.15** (0.494)	-2.86*** (0.452)	-3.56*** (0.356)	3.06*** (0.191)	3.11*** (0.192)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Rating FE	Yes	Yes	Yes	Yes	Yes	Yes
Issuer country FE	No	Yes	No	Yes	No	Yes
N	3,884	3,883	3,884	3,883	3,884	3,883
R ²	.199	.476	.437	.640	.243	.253

B. Bank-level analysis

	log(value outstanding _{b,t})			
	(1)	(2)	(3)	(4)
HaircutGap _{b,t-1}	0.27*** (0.048)	0.77*** (0.055)	0.58*** (0.087)	0.54*** (0.092)
Time FE	No	Yes	Yes	Yes
Issuing bank FE	Yes	Yes	Yes	Yes
Rating group FE	No	No	Yes	—
Rating FE	No	No	No	Yes
N	25,212	25,212	23,327	20,599
R ²	.954	.955	.955	.963

This table presents the coefficients from the regressions related to the issuance of bank bonds. Panel A presents estimates of Equation (9) at the granular security-time level. In columns 1 and 2, the dependent variable denotes the log value of security s issued at time t . In columns 3 and 4, the outcome variable $\text{Spread}_{s,t}$ is the spread between yield-to-maturity of the issued bond and the AAA Euro Area yield of a matching maturity. The outcome variable in columns 5 and 6 is the log of maturity of the newly issued bond s issued at its issuance time t . We regress the outcome variable on the haircut gap measure of the security. Because the security is issued only at time t , previous levels of the haircut gap cannot be observed. We address this issue by constructing an expected value of the haircut gap. We use publicly available ECB haircut policy rules to compute ECB haircuts. We employ the machine learning algorithm to predict the private market haircut using the issuer rating as of $t-1$ along with the actual maturity, asset type and coupon type of the security s . The control variables include the spread and maturity when the quantity issued is the outcome variable, the quantity and maturity when the spread is the dependent variable, and the spread and quantity for the bond maturity analysis. Panel B presents estimates of Equation (10), which collapses the data at the bank-time level. This strategy allows us to distinguish between new bond issuances, which simply replace maturing debt and issuances that increase the bank’s overall borrowing

(a) Table 8: bank-bond issuance.

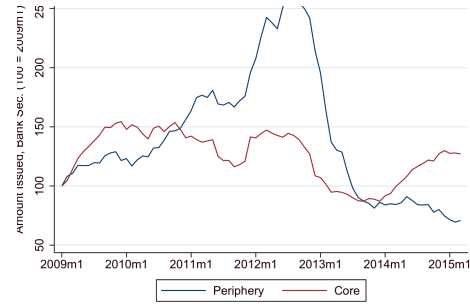


Figure 6
Issuance of bank bonds

(b) Figure 6: issuance of bank bonds.

5.7 Table 9: balance-sheet effects

Read:

- For issuing banks, a higher haircut gap raises bonds issued and total liabilities.
- It does not raise loans from banks or deposits, so the bank-bond channel does not simply substitute for other liabilities.

- A one-standard-deviation increase in haircut gap corresponds to about a 13% increase in bonds issued and a 2% increase in total liabilities.
- For pledging banks, high-gap pledged bonds are associated with more ECB liquidity and larger bank-bond portfolios.
- A one-standard-deviation increase corresponds to a 19% increase in ECB liquidity and a 13% increase in bank bonds held.

Economic message. The balance-sheet evidence shows that the channel expands bank-bond interconnect-
edness on net rather than replacing interbank loans one-for-one.

A. Issuing banks					
	log (bonds issued) $_{b,t}$	log (loans from banks) $_{b,t}$	log (deposits) $_{b,t}$	log (total liabilities) $_{b,t}$	(debt/equity) $_{b,t}$
	(1)	(2)	(3)	(4)	(5)
HaircutGap (of issued bonds) $_{b,t-1}$	0.918*** (0.143)	-0.0868 (0.0906)	-0.064 (0.0529)	0.137*** (0.0264)	3.963*** (1.019)
Time FE	Yes	Yes	Yes	Yes	Yes
Issuing bank FE	Yes	Yes	Yes	Yes	Yes
N	7,371	6,488	6,882	6,997	6,997
R ²	.933	.906	.899	.972	.852
B. Pledging banks					
	log(ECB liquidity) $_{b,t}$	log(bank bonds held) $_{b,t}$		log(lending to banks) $_{b,t}$	
	(1)	(2)	(3)		
HaircutGap (of pledged bonds) $_{b,t-1}$	1.339*** (0.114)	0.862*** (0.0734)	-0.0117 (0.158)		
Time FE	Yes	Yes	Yes		
Pledging bank FE	Yes	Yes	Yes		
N	6,350	10,246	8,829		
R ²	.739	.915	.805		
This table presents the coefficients from the balances sheet analysis of issuing banks (panel A) and pledging banks (panel B). The haircut gap variable is computed as a weighted average of issued (pledged) bank bonds for the issuing (pledging) banks. Standard errors are two-way clustered at the bank and time level in parentheses.					
*** $p<.01$; ** $p<.05$; * $p<.1$.					

Figure 6: Table 9: balance-sheet analysis of issuing and pledging banks.

5.8 Tables 10–11: systemic-risk implications

Read: Table 10:

- Haircut gaps increase correlation in bond prices between pledging banks and issuing banks.
- The effect is positive for domestic links and negative for foreign links.
- The correlation effect is stronger for systemically important issuing banks.
- It is also stronger during low market-return periods.
- A one-standard-deviation higher haircut gap is associated with a 7.3% stronger bond-price correlation in low-return periods and 14.1% for domestic systemically important issuing banks.

Read: Table 11:

- Haircut gaps increase bank bond-price correlations especially during periods of sovereign stress.
- The sovereign-yield interaction is positive whether sovereign stress is measured as a top-20% dummy or as a continuous yield variable.
- Instrumented pledging specifications deliver consistent conclusions.

Economic message. The final tables connect the collateral subsidy to systemic risk. The LOLR-induced network becomes most dangerous in bad states, exactly when correlated bank losses matter most.

Table 10
Haircut gap and correlation in market outcomes between pledging and issuing banks

	CorrBond prices _(i,j)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
HaircutGap _(i,j-1) × Domestic _(i,j)	0.416*** (0.0752)	0.328*** (0.0572)	0.262*** (0.0818)	0.331*** (0.0773)				
HaircutGap _(i,j-1) × Foreign _(i,j)	-0.316*** (0.0452)	-0.375*** (0.0441)						
HaircutGap _(i,j-1) × Domestic _(i,j) × SRSK _(i,j-1)			0.689*** (0.121)	0.579*** (0.102)				
HaircutGap _(i,j-1) × SRSK _(i,j-1)			-0.213*** (0.0680)	-0.222*** (0.0690)				
HaircutGap _(i,j-1)			-0.796*** (0.0508)	-0.827*** (0.0464)				
HaircutGap _(i,j-1) × Normal returns _(i,j-1)					0.0734 (0.0489)	0.0125 (0.0396)		
HaircutGap _(i,j-1) × Low returns _(i,j-1)					0.243*** (0.0939)	0.138*** (0.0382)		
Value pledged _(i,j)							0.0313*** (0.0061)	0.0267*** (0.0045)
Value outstanding _(i,j)								
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pledging bank-issuing bank FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country-Time FE	No	Yes	No	Yes	No	Yes	No	Yes
N	117,108	117,108	116,359	116,340	117,108	117,108	117,108	117,108
R ²	.638	.652	.639	.653	.636	.650	.637	.651

This table presents the coefficients from the analysis of correlation of market outcomes between pledging and issuing banks, as described by Equation (11). The table reports the main coefficients of interest but the regressions estimate all coefficients, including remaining interaction terms and controls. A security is considered to be Domestic if the pledging and issuing banks have the same country of location, and as Foreign otherwise. SRSK takes the value of one if the issuing bank's SRSK ratio is in the top 10% of the SRSK ratios of all banks in the respective country, and zero otherwise. Returns are denoted as Low in periods when the average bond returns are in the both 20th percentile of all bond returns across time and as Normal otherwise. Columns 7 and 8 show the standardized coefficients of the analysis of the impact of the haircut gap on the bond price correlation directly through its effects on the security pledging. The first-stage instrument is the amount of security pledged with the haircut gap using specification from Equation (4). The fitted values from the first stage are used as the independent variable to estimate the effects of increased pledging (and holdings) of bank bonds on bond price correlation columns 7 and 8). Controls include lagged correlation between bond prices of banks and correlations between bank bonds and domestic sovereign bonds. The dash refers to the fact that the fixed effects are not applicable as they are nested in different (higher-order) fixed effects. Standard errors are clustered at the issuing bank-pledging bank and time levels in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 11
Haircut gaps, bonds holdings and sovereign stress

	CorrBond prices _(i,j)							
	Sovereign yield (dummy, top 20%)				Sovereign yield (cont. variable)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
HaircutGap _(i,j-1)	0.151** (0.0591)	0.151** (0.0584)	0.142** (0.0590)				0.0254*** (0.00548)	0.0256*** (0.00548)
HaircutGap _(i,j-1) × Sovereign Yield _(i,j-1)	0.0044 (0.0407)	0.00019 (0.0404)	0.00776 (0.0407)				-0.0325 (0.0621)	-0.0332 (0.0615)
Value pledged _(i,j)				0.0393*** (0.00731)	0.0395*** (0.00732)	0.0383*** (0.00730)		0.0277*** (0.00541)
Value outstanding _(i,j)								
Value pledged _(i,j) × Sovereign Yield _(i,j-1)				0.00845 (0.00517)	0.00810 (0.00510)	0.00828 (0.00517)		
Controls	Yes	Yes ^a	Yes ^b	Yes	Yes ^a	Yes ^b	Yes	Yes ^a
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pledging bank-issuing bank FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	117,108	117,108	117,108	117,108	117,108	117,108	117,108	117,108
R ²	.638	.638	.638	.640	.640	.640	.636	.636

This table presents the coefficients from the analysis of correlation of bond prices between pledging and issuing bank, as described by Equation (11). In columns 1-4, the Sovereign Yield_(i,j-1) is a dummy variable that takes the value of one if the yield on the 10-year sovereign bond issued by the domestic government of the issuing bank belongs to the top 20% of the 10-year sovereign bond distribution, and zero otherwise. In columns 5-9 Sovereign Yield_(i,j-1) is a continuous variable of the 10-year sovereign bond issued by the domestic government of the issuing bank. Columns 4-6 present the analysis of the impact of the haircut gap on the bond price correlation directly through its effects on the security pledging. The first-stage instrument is the amount of security pledged with the haircut gap using specification from Equation (4). The fitted values from the first stage are used as the independent variable to estimate the effects of increased pledging (and holdings) of bank bonds on bond price correlation. Coefficients reported in columns 4-6 are standardized. All specifications control for lagged correlation between bond prices of banks and correlation between bank bonds and domestic sovereign bonds. In addition, we consider the following controls denoted as *a*: an interaction of the share of sovereign bond pledged by issuing bank to the total value outstanding of the sovereign bonds and the respective sovereign bond yield, and an interaction of the share of sovereign bond pledged by pledging bank to the total value outstanding of the sovereign bonds and the respective sovereign bond yield. Alternatively, we also include interaction of haircut gap and low return period as a control variable (denoted as *b*). Standard errors are clustered at the issuing bank-pledging bank and time levels in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6 Short summary

- The paper studies whether LOLR policy changes bank interconnectedness.
- The central variable is the haircut gap: private repo haircut minus ECB haircut.
- A high haircut gap makes a bond attractive as central-bank collateral.
- The data link security-level pledging, holdings, haircuts, ratings, bank balance sheets, and prices.
- Banks pledge more high-gap securities to the ECB.
- The effect is stronger for bank bonds than for government bonds.
- Within bank bonds, the effect is strongest for domestic bonds, periphery banks, and systemically important issuing banks.
- ECB institutional haircut rules support the causal interpretation.
- Banks also hold more high-gap bank bonds; nonbanks without LOLR access do not.
- High expected haircut gaps stimulate bank-bond issuance, lower spreads, and lengthen maturities.
- Issuing banks expand liabilities and leverage; pledging banks increase ECB liquidity and bank-bond holdings.
- The increase in bank-bond linkages is not offset by lower interbank lending.
- Bond-price correlations between linked banks rise, especially during low-return periods and sovereign stress.
- The policy trade-off is clear: LOLR support provides liquidity insurance but can increase systemic risk through collateral incentives.

7 Conclusion

7.1 Core conclusion

Jasova, Laeven, Mendicino, Peydró, and Supera show that LOLR policy can increase bank interconnectedness through the haircut-gap channel. When ECB haircuts are more favorable than private-market haircuts, banks have incentives to buy and pledge those high-gap securities. This effect is especially strong for domestic bank bonds and for bonds issued by systemically important banks.

7.2 Economic intuition

The ECB provides liquidity against collateral. If the ECB values a bond more favorably than the private repo market, the bond becomes a cheap way to access central-bank funding. Banks with LOLR access respond by holding and pledging those bonds. When the bonds are issued by other banks, this creates direct bank-to-bank balance-sheet links. In good times, this may support funding; in bad times, it raises correlated exposure.

7.3 Takeaway

The paper's main contribution is to show that central-bank collateral policy has systemic-risk consequences. Haircuts are not merely technical parameters in liquidity operations. They shape banks' portfolio choices, issuance incentives, balance-sheet expansion, and market comovement. LOLR policy can be stabilizing in the short run while increasing interconnectedness and systemic risk at the margin.